LAB - 01

Experiment -

Study of LAN Topology, Network Devices and Address configuration.

Problem Statement -

1. Draw the Network topology :-

- Identify and draw the LAN topology of your organization (e.g., Star, Bus, Mesh).
- Clearly label all components such as Router, Switch, Access point and end devices (PCs, laptops etc)

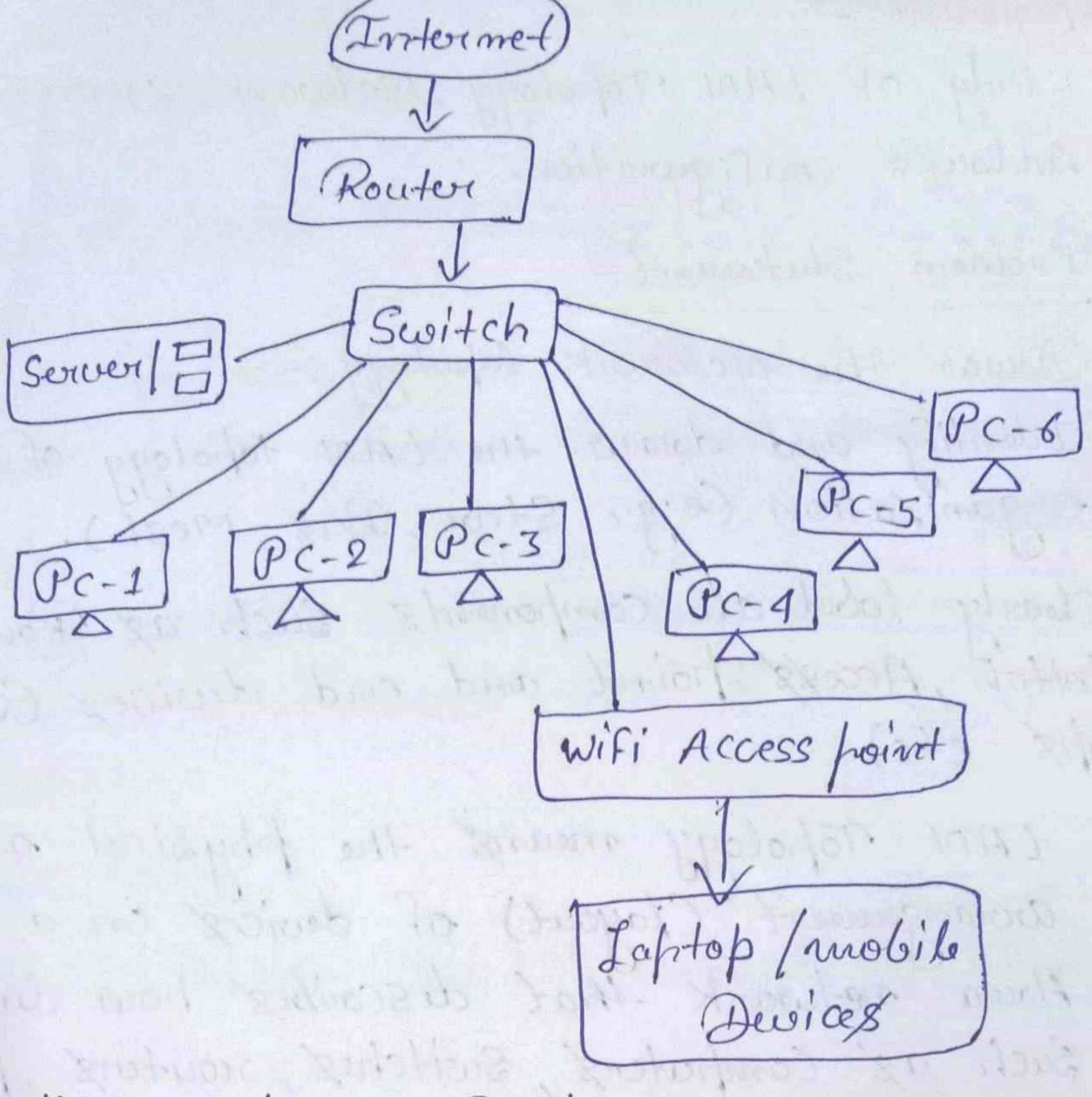
Ans :- LAN Topology means the physical or logical arrangement (layout) of devices in a local Area network that describes how devices such as computers, switches, routers, printers and access points are interconnected.

Example :- college labs mainly uses a Star Topology.

- Star topology - All devices are connected to a central Switch or hub.

Now,

Diagram of network topology -



2. Identify Network Devices -
   - List all Networking devices used in the LAN.
   - Mention the Function of each device.

Ans - There are following networking devices use in LAN :-

i) Router
ii) Switch
iii) Access point
iv) Server

v) Desktop computers
vi) Laptop / mobile devices

Functions of listed devices :-

i) Router :-

- Connects the college LAN to the Internet and routes data between different networks.

ii) Switch :-

- Connects multiple computers and devices within the LAN and forwards data to the correct destination.

iii) Access point :-

- Provides wireless (wi-fi) connectivity to laptops and mobile devices.

iv) Server :-

- Stores files, applications and user data and provides network services to clients.

v) Desktop computers :-

- End devices used by students and faculty for practical work and accessing network resources.

vi) Laptop / Mobile devices :-
- wireless client devices that access the network through the Access point.

3) Analyze connection Types :-
- Specify whether connections are :-
  - Wired (Ethernet)
  - wireless (wi-fi)
- mention where each type is used.

Ans :- Connection Types -

1. Wired Connections (Ethernet) :-
   - medium - Ethernet (LAN) cable.
   - Used for - Desktop computers, Servers, network switch and router.
   - Reason - provides high speed, stable connection and low latency.

2. wireless connection (wi-fi) :-
   - medium - wi-fi.
   - Used for - Laptops, Smartphones and other wireless devices through the Access point.
   - Reason - Provides mobility and allows users to connect without cables.

Q 4) Measure Network Speed:

- Find the link Speed of:
  - wired network (if available)
  - wireless network
- Mention the tools / commands used.

Ans :- Tool used -

- windows Task Manager.

Command used -

- none.

1) wired network (not available)

2) wireless network

- measured Speed - 72.0 Kbps.

Q 5) Find System Network Details :-

- Determine and record the following of your Computer :-
  - MAC Address
  - IP Address (IPv4)
  - Subnet Mask
- Mention the Command (s) used.

Ans :- Command used - ipconfig /all

- MAC Address - 90-65-84-93-E7-4D
- IP Address - 10.85.23.124
- Subnet - Mask - 255.255.248.0

Q6) Explain observations:-

- Briefly explain:-
  - Difference b/w wired and wireless speed.
  - Role of subnet mask in networking.

Ans:- There are following differences:-

1. Wired Network (Ethernet):-

- A wired network transfers data through Ethernet cables.
- It provides higher speed and more stable connectivity.
- It has lower latency and fewer chances of interferences.
- It is commonly used in computer labs, offices and servers.

2. Wireless Network (wi-fi):-

- A wireless network transfers data using rac waves.
- Its speed depends on signal strength, dista ce and interference.
- It provides mobility and allows devices to connect without cables.

Role of Subnet Mask in Networking-

A Subnet Mask is used to divide an IP address into two parts:-

i) Network part
ii) Host part

- It identifies the network and host portion of an IP address.
- It helps devices communicate within the correct network.
- It helps routers determine where to send data packets.

Ex:- IP Address : 172.25.71.217
Subnet Mask : 255.255.240.0

- It helps identify the network to which the device belongs.